Practical-7

Aim: **To solve queries using the concept of sub query.**

THEORY :

- A subquery is a query within another query. The outer query is called as main query and inner query is called as subquery.
- The subquery generally executes first when the subquery doesn't have any co-relation with the main query.
- Syntax:

```
SELECT column_name
FROM table_name
WHERE column_name expression operator
(SELECT COLUMN_NAME  from TABLE_NAME   WHERE ... );
```

Q-1) Write a query to display the last name and hire date of any employee in the same department as smith. Exclude smith

Ans: select l_name,hiredate from employee where dept_name=(select dept_name from employee where emp_name='smith') and emp_name <> 'smith';

```
SQL> select l_name,hiredate from employee where dept_name=( select dept_name from employee where emp_name='smith') and emp_name <> 'smith';

L_NAME                         HIREDATE
------------------------------ ---------
wales                          30-NOV-95
```

Q-2) Give name of customers who are depositors having same branch city of mr. sunil.

Ans: select cname from deposit where bname=(select bname from deposit where cname='sunil');

```
SQL> select cname from deposit where bname=(select bname from deposit where cname='sunil');

CNAME
---------------
sunil
vinit
```

Q-3) Give deposit details and loan details of customer in same city where pramod is living.

Ans: Select d.a_no,d.cname,d.bname,d.amount,d.a_date,b.loanno,b.cname,b.amount from deposit d jion borrow b on d.bname=b.bname where d.bname in(select bname from deposit where cname='pramod')And b.bname in(select bname from deposit where cname='pramod');

```
SQL> select d.a_no, d.cname, d.bname, d.amount, d.a_date,b.loanno,b.cname,b.bname,b.amount from deposit d join borrow b on d.bname=b.bname
  2   where d.bname in(select bname from deposit where cname='pramod') and b.bname in(select bname from deposit where cname='pramod');

A_NO  CNAME          BNAME          AMOUNT A_DATE    LOANN CNAME          BNAME          AMOUNT
----- -------------- ---------- ---------- -------- ----- -------------- ---------- ----------
110   pramod         virar           12000 14-JAN-06 208   pramod         virar           12000
102   sunil          virar            5500 15-JUL-06 208   pramod         virar           12000
```

Q-4)	Create a query to display the employee numbers and last names of all employees who earn more than the average salary. Sort the results in ascending order of salary.
Ans:	select emp_no,l_name from employee where emp_sal > (select AVG(emp_sal) from employee) order by emp_sal; SQL> select emp_no,l_name from employee where emp_sal > (select AVG(emp_sal) from employee) order by emp_sal; EMP_NO L_NAME --- --- 106 joseph 107 jha 104 sharma 105 patel
Q-5)	Give names of depositors having same living city as mr. anil and having deposit amount greater than 2000
Ans:	select * from deposit where bname=(select bname from borrow where cname='anil') and amount>2000; SQL> select * from deposit where bname=(select bname from borrow where cname='anil') and amount>2000; A_NO CNAME BNAME AMOUNT A_DATE --- --- --- --- --- 113 neel karolbagh 5500 16-JAN-06
Q-6)	Display the last name and salary of every employee who reports to ford.
Ans:	select l_name from employee where manager_id = (select emp_no from employee where emp_name = 'Anita'); SQL> select l_name from employee where manager_id = (select emp_no from employee where emp_name = 'Anita'); L_NAME --- shah wales jodeph
Q-7)	Display the department number, name, and job for every employee in the accounting department.
Ans:	select dept_no, emp_name, job_id from employee where job_id = (select job_id from job where job_title like 'Account'); SQL> select dept_no, emp_name, job_id from employee where job_id = (select job_id from job where job_title like 'Account'); DEPT_NO EMP_NAME JOB_ID --- --- --- 10 sneha fi_acc
Q-8)	List the name of branch having highest number of depositors.

Ans: select bname from deposit group by bname having count(bname) = (select max(count (bname)) from deposit group by bname);

```
SQL> select bname from deposit group by bname having count(bname) = (select max(count (bname)) from deposit group by bname);

BNAME
----------
bombay
virar
andheri
```

Q-9) Give the name of cities where in which the maximum numbers of branches are located.

Ans: select bname from deposit group by bname having count(bname) = (select max(count (bname)) from deposit group by bname);

```
SQL> select bname from deposit group by bname having count(bname) = (select max(count (bname)) from deposit group by bname);

BNAME
----------
bombay
virar
andheri
```

Q-10) Give name of customers living in same city where maximum depositors are located.

Ans: select cname,city from customer where city IN(select city from customer group by city having count(city) > 1 AND city IN(select bname from deposit group by bname having count(bname) = (select max(count (bname)) from deposit group by bname)));

```
SQL> select cname,city from customer where city IN(select city from customer group by city having count(city) > 1 AND city IN(select bname from deposit group by bname having count(bname) = (select max(count (bname)) from deposit group
 by bname)));

no rows selected
```

Job schema:

create table job(job_id varchar2(15),job_title Varchar2(30),min_sal Number(7,2),max_sal Number(7,2));

```
SQL> desc job;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------
 JOB_ID                                             VARCHAR2(15)
 JOB_TITLE                                          VARCHAR2(30)
 MIN_SAL                                            NUMBER(7,2)
 MAX_SAL                                            NUMBER(7,2)
```

insert into job values('&job_id','&job_title',&min_sal,&max_sal);

```
SQL> select * from job;

JOB_ID          JOB_TITLE                        MIN_SAL    MAX_SAL
--------------- ------------------------------ ---------- ----------
it_prog         Programmer                           4000      10000
mk_mgr          Marketing manager                    9000      15000
fi_mgr          Finance manager                      8200      12000
fi_acc          Account                              4200       9000
lec             Lecturer                             6000      17000
comp_op         Computer Operator                    1500       3000
```

Employee schema:

create table employee(emp_no Number(3),emp_name Varchar2(30),emp_sal Number(8,2),emp_comm Number(6,1),dept_no Number(3),l_name Varchar2(30),dept_name Varchar2(30),job_id Varchar2(15),location Varchar2(15),manager_id Number(5),hiredate Date);

```
SQL> desc employee;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------
 EMP_NO                                             NUMBER(3)
 EMP_NAME                                           VARCHAR2(30)
 EMP_SAL                                            NUMBER(8,2)
 EMP_COMM                                           NUMBER(6,1)
 DEPT_NO                                            NUMBER(3)
 L_NAME                                             VARCHAR2(30)
 DEPT_NAME                                          VARCHAR2(30)
 JOB_ID                                             VARCHAR2(15)
 LOCATION                                           VARCHAR2(15)
 MANAGER_ID                                         NUMBER(5)
 HIREDATE                                           DATE
```

insert into employee values(&emp_no,'&emp_name',&emp_sal,&emp_comm,&dept_no,'&l_name', '&dept_name','&job_id','&location',&manager_id,'&hiredate');

```
SQL> select * from employee;

    EMP_NO EMP_NAME                       EMP_SAL   EMP_COMM    DEPT_NO L_NAME                       DEPT_NAME                    JOB_ID         LOCATION    MANAGER_ID HIREDATE
---------- ------------------------------ ---------- ---------- ---------- ------------------------------ ------------------------------ --------------- --------------- ---------- -----
       101 smith                                 800                    20 shah                           machin learning                fi_mgr         toronto            105 09-AUG-96
       102 Snehal                               1600        300         25 gupta                          data science                   lec            las vegas              14-MAR-96
       103 adama                                1100          0         20 wales                          machin learning                mk_mgr         ontario            105 30-NOV-95
       104 aman                                 3000                    15 sharma                         virtual reality                comp_op        mexico              12 02-OCT-97
       105 anita                                5000      50000         10 patel                          big data analytics             comp_op        germany            107 01-JAN-98
       106 sneha                                2450      24500         10 joseph                         big data analytics             fi_acc         melbourne          105 26-SEP-97
       107 anamika                              2975                    30 jha                            artificial intelligence        it_prog        new yprk               15-JUL-97

7 rows selected.
```

Deposit schema:

create table deposit(a_no Varchar2(5),cname Varchar2(15),bname Varchar2(10),amount Number(7,2),a_date Date);

```
SQL> desc deposit;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------
 A_NO                                               VARCHAR2(5)
 CNAME                                              VARCHAR2(15)
 BNAME                                              VARCHAR2(10)
 AMOUNT                                             NUMBER(7,2)
 A_DATE                                             DATE
```

insert into deposit values(&a_no,'&cname','&bname',,&amount,'&a_date');

```
SQL> select * from deposit;

A_NO  CNAME           BNAME         AMOUNT A_DATE
----- --------------- --------- ---------- ---------
107   om              nagpur          4500 28-AUG-06
101   Anil            andheri         7000 01-JAN-06
102   sunil           virar           5000 15-JUL-06
103   jay             villeparle      6500 12-MAR-06
104   vijay           andheri         8000 17-SEP-06
105   keyur           dadar           7500 19-NOV-06
106   mayur           borivali        5500 21-DEC-06
108   Raj             bombay          9500 21-AUG-06
109   jenil           bombay          6500 20-JAN-06
110   savan           bombay          3500 15-MAR-06

10 rows selected.
```

Borrow schema:

create table borrow(loanno Varchar2(5),cname Varchar2(15),bname Varchar2(10),amount number(10));

```
SQL> desc borrow;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------
 LOANNO                                             VARCHAR2(5)
 CNAME                                              VARCHAR2(15)
 BNAME                                              VARCHAR2(10)
 AMOUNT                                             NUMBER(10)
```

insert into borrow values(&loann,'&cname','&bname',&amount);

```
SQL> select * from borrow;

LOANN CNAME           BNAME          AMOUNT
----- --------------- ---------- ----------
208   om              nagpur           6500
201   anil            karolbagh        1000
206   mehul           vrce             4000
```

Conclusion:

In this practical we learn about this concept of sub query.

Student sign: Faculty sign: Marks: